Foils/wraps: HMTc Standard

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CORRESPONDENCE

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for foils/wraps in Food-Contact Consumer Goods and Kitchenware, evaluated on specific element migration INTO FOOD OR A FOOD SIMULANT from the polymeric food-contact surface, µg/kg of food, determined by the Reg (EU) 10/2011 Annex III to V simulant, temperature and contact-time conventions. This is a MIGRATION basis and not a total-content basis: no cell on this row limits how much of an element the material CONTAINS, and no cell on this row is expressed per kg of material.. 6 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum for this exact product form expressed as specific element migration into food or a food simulant; a disclosed read-across from the nearest applicable sovereign value in the same basis where the food-contact material family is expressly excluded from or not reached by that instrument; a substantiated not-material determination; or a control. No cell is occurrence-derived, and no cell borrows a value across a basis.

2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	control	measured every lot; no human-food maximum

Cadmium (Cd)	Tier 1	control	measured every lot; no human-food maximum
Total arsenic (tAs)	Tier 2	control	measured every lot; no human-food maximum
Inorganic arsenic (iAs)	Tier 1	control	measured every lot; no human-food maximum
Mercury, total (tHg)	Tier 1	control	measured every lot; no human-food maximum
Methylmercury (MeHg)	Tier 1	n/m	not material in this matrix
Nickel (Ni)	Tier 2	control	measured every lot; no human-food maximum
Tin (Sn)	Tier 2	control	measured every lot; no human-food maximum
Aluminium (Al)	Tier 2	control	measured every lot; no human-food maximum
Total chromium (Cr)	Tier 2	control	measured every lot; no human-food maximum
Hexavalent chromium (Cr-VI)	Tier 2	control	measured every lot; no human-food maximum

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION

Lead (Pb)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Lead on this row arrives from printing inks and surface coatings on film and foil, from stabilisers and pigments in PVC cling compound, and from the alloy and rolling-lubricant residues of the foil itself. The material-content instruments (US CPSIA total lead in children's-product substrate, and Japan Notice 370 and Korea MFDS Notice 2024-29 percent-by-weight lead in food-contact metals, solder and tin plating) are separate obligations in a different basis and are neither this cell nor its companion.

Cadmium (Cd)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Cadmium on this row arrives from cadmium-pigmented printing and from PVC cling compounds. The EU REACH Annex XVII cadmium restrictions on plastics, paints and recovered PVC are percent-by-weight CONTENT restrictions and are not read into this release cell.

Total arsenic (tAs)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Arsenic is determined as the TOTAL element and cannot be satisfied by an inorganic-arsenic determination. It arises as a trace impurity of pigments, fillers and smelter-route metal.

Inorganic arsenic (iAs)	control	Arsenic released from a polymeric film or from a rolled-metal surface is inorganic by construction, so inorganic arsenic cannot be bounded BELOW the total-arsenic result. This row's total-arsenic cell is itself a control with no published parent, so a speciation-bounded not-material determination is unavailable and the fail-safe applies. This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. The film configuration's total-arsenic companion (eu_10_-2011_tas_sml_nd, total arsenic not detectable at 0.01 mg/kg food) bounds the inorganic fraction on that configuration only, and is disclosed as a companion rather than published as this cell's ceiling.

Mercury, total (tHg)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Mercury has no functional role in either configuration and appears only as a trace raw-material impurity, but it remains a mandatory panel determination on every certified lot.

Methylmercury (MeHg)	n/m	Methylmercury is formed by microbial methylation in aquatic food chains, where an anaerobic sulfate- or iron-reducing community acts on a bioavailable inorganic-mercury pool over long residence times. Neither configuration of this row supplies that: a cured polymer film and a rolled aluminium foil contain no organic carbon available to a methylating community, contact is short, aerobic and dry-to-moist rather than submerged and anoxic, and neither material holds a mercury pool to methylate. No credible sovereign sets a methylmercury requirement for any food-contact material in any basis. Total mercury remains a mandatory control on this row for both configurations, and on the polymeric-film configuration the Reg (EU) 10/2011 total-mercury non-detect companion additionally bounds any organomercury fraction on that configuration. Methylmercury is not material and never substitutes for the total-mercury cell.

Nickel (Ni)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Nickel is the clearest case for the carve-out on this row. The plastics nickel specific migration limit of 0.02 mg/kg food was written for a polymer surface; it is not the achievable release figure for a bare rolled-metal surface, so importing it as this row's published ceiling would publish an unfounded number rather than a strict one. It binds the film configuration as a companion, where it is both applicable and achievable.

Tin (Sn)	control	Regulation (EU) 10/2011 Annex II lists no tin entry, and no credible sovereign sets a tin migration or release ceiling for a food-contact article of any material family. Tin is nonetheless a mandatory panel cell, because tin stabilisers in PVC compounds and tin-bearing solder are live pathways in parts of this category, so it is tested on every certified lot with no invented ceiling. This cell carries NO companion: the only sovereign tin numbers in existence are maximum levels on canned food, which are a FOOD basis on the contents of a tinplate container, and they are neither applicable to this row nor publishable as a migration ceiling. They are named as a companion only on the rows where a tinplate or crown component actually exists. On this row that conclusion is unchanged by the move of the rest of the panel to the metal-article release control: tin has no entry in the plastics instrument, so it never had a companion here and does not acquire one.

Aluminium (Al)	control	Aluminium on this row is a control, and the carve-out is deliberate. The polymeric-film configuration is bound by the EU 10/2011 Annex II aluminium specific migration limit of 1 mg/kg food, which is disclosed as this cell's companion and is verified on every film lot. The ALUMINIUM-FOIL configuration is not: rolled aluminium foil is a metal article whose food-contact surface is aluminium itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign sets an aluminium release ceiling for a metal foil. Publishing the plastics SML against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading, and it is documented that aluminium foil in contact with acidic or salted food exceeds it. HMC therefore mandates release testing of the foil configuration under the declared contact conditions with no invented ceiling, and reports the film and foil configurations separately rather than pooling them.
Total chromium (Cr)	control	This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED-ALUMINIUM-FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled. Total chromium is retained as a mandatory determination

Hexavalent chromium (Cr-VI)	control	Hexavalent chromium is the health-relevant chromium species and it cannot be dismissed on this row: chromate-conversion and passivation treatments on metal surfaces, and chrome-pigmented printing on either configuration, are live pathways. Reg (EU) 10/2011 Annex II point 2 controls it on the FILM configuration through the total-chromium non-detect requirement, which is disclosed as that configuration's companion and which HMTc holds without the 3.6 mg/kg food relaxation. On the FOIL configuration no sovereign release ceiling exists and this row's total-chromium cell is itself a control, so no published parent bounds the species and the fail-safe applies. This taxonomy row covers two material configurations and the published cell governs both, so it is a control. The polymeric cling and barrier-film configuration is bound by the Reg (EU) 10/2011 Annex II value disclosed as this cell's COMPANION, which is a mandatory obligation verified on every film lot. The ROLLED ALUMINIUM FOIL configuration is not bound by it: foil is a metal article whose food-contact surface is the element itself, metals and alloys are a Regulation 1935/2004 Annex I material family for which the EU has adopted no specific measure, and no credible sovereign - EU, UK, Codex, US, AU-NZ, Canada or Hong Kong - sets an element release ceiling for a metal foil. Publishing the plastics value against aluminium foil would take a number written for a polymer surface and assert it of an article made of the element, which is a misapplication rather than a conservative reading. That is the reasoning this row already applied to aluminium, and it holds identically for every other element on the panel, so the whole panel takes the metal-article release control used on cookware-metal--alloy and utensils-metal rather than a plastics anchor. Testing is mandatory on every certified lot with no invented ceiling, and the film and foil configurations are reported separately and never pooled.

4 | Cite this standard

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/food--packaging-foils-wraps. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

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